

Climate Shocks, Health Finance, and Economic Outcomes

Main Results Recap + April 2026 Committee Feedback Response

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Roadmap

Part 1: where we were before April. Part 2: what changed in response to committee feedback.

Part 1 — Main Results Recap

- Project setup and empirical specification
- State-level FE: cold and heat drive opposite financial channels
- County-level FE: cold raises debt + premiums; heat depresses income
- Weather-swing (delta) analysis: AQI ratchet, drought-PCPI loss
- Cross-level synthesis

Part 2 — Committee Feedback Response

- Random-effects robustness (Hausman)
- Post-exit local-projection dynamics
- DiD with never-exposed counties (2012 drought; Callaway-Sant'Anna)
- Propagation pathways: literature + descriptives
- Humidity (PRISM): parked status
- What changed substantively; open caveats

Part 1 — Main Results Recap

Project setup, state, county, weather-swing

Project Background

Climate, health finance, and economic outcomes in the U.S., 2011–2023

Research Question

- Do climate shocks (cold, heat, drought, air quality) affect the **cost and accessibility** of health care?
- Do they propagate into **household economic outcomes** — income, debt, employment?

Motivation

- Climate adaptation policy needs **health-finance channel** evidence
- Literature centres on mortality; **cost & debt channels understudied**
- U.S. two-way FE enables within-unit identification at state and county

Data Coverage

Domain	Source
Climate (Temp, HDD, CDD)	NOAA/NCEI
Drought index (PDSI)	NOAA/NCEI
Air quality (AQI)	EPA
HIX premiums	HIX Compare
Medical debt	Urban Institute
Hospital bad debt	NASHP HCT
Employer insurance	AHRQ MEPS-IC
Public payers	CMS NHE
HH income / employment	ACS / BEA

Panel: 2011–2023; ~3,155 counties;

state panel: 51 states × 13 years.

Empirical Specification

Two-way fixed effects with distributed lags; complementary LP framework for dynamics; dollar outcomes inflation-adjusted to \$2023

Dynamic Panel / Distributed Lag (DP/DL)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^2 \beta_h \text{Shock}_{i,t-h} + \mathbf{X}'_{it} \delta + \varepsilon_{it}$$

- Main FE estimator for contemporaneous and lagged effects
- α_i unit FE (State or fips); γ_t year FE
- SE clustered at **state level**; RA cluster for premiums

Shocks: High_CDD, High_HDD, Z_Temp, Z_Precip, PDSI, High_AQI_Max (>100). **Outcomes:** employer premium contribution (MEPS-IC), benchmark silver premium (HIX), medical debt share/median (Urban), hospital bad debt per capita (NASHP), total/Medicaid/Medicare per-enrollee spending (CMS), HH income, employment, PCPI (ACS/BEA).

Local Projections (LP, Jordà 2005)

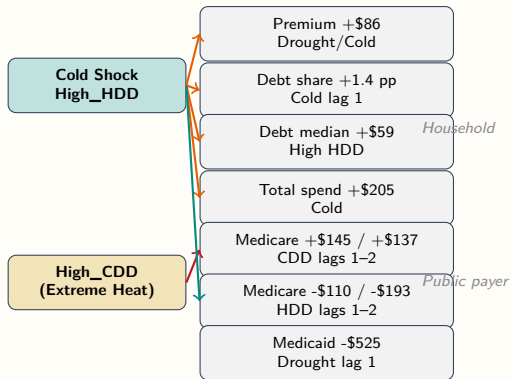
$$Y_{i,t+h} = \alpha_i^h + \gamma_t^h + \beta_h \text{Shock}_{i,t} + \mathbf{X}'_{it} \delta_h + \varepsilon_{i,t+h}$$

- Separate regression per horizon
 $h \in \{-2, \dots, +3\}$
- Negative horizons = placebo pre-trend checks
- More flexible if lag shapes are misspecified

State-Level Findings

Cold raises premiums and debt; drought hits Medicaid and PCPI;
heat and cold move public payers in opposite directions

State-Level: Cold and Heat Drive Opposite Financial Channels



Cold channel:

- Premiums, debt, and total spending rise
- Medicare falls with lagged HDD exposure

Heat channel (CDD):

- Medicare rises 1–2 years later

Drought:

- Medicaid falls and premiums rise

Heat and cold affect public payers in opposite directions — novel finding.

State: Headline Coefficients

Selected significant lags; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Household outcomes

Coefficient (state-clustered)	Estimate	
Employer Prem.: Severe Drought ($h=0$)	+\$86.4	***
Employer Prem.: Severe Drought (lag 1)	+\$65.1	**
Employer Prem.: Pct PM _{2.5} days	+\$20.8	***
Debt Share: Cold Shock (lag 1)	+0.0135	**
Debt Median: High HDD ($h=0$)	+\$58.9	**
Debt Median: Heat Shock (lag 2)	+\$47.7	**
Total PC Health Exp: Cold Shock ($h=0$)	+\$205.3	**

Public payer outcomes

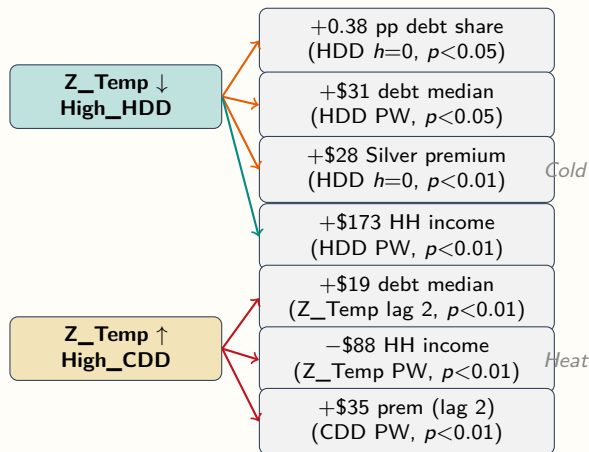
Coefficient (state-clustered)	Estimate	
Medicaid PE: Extreme Drought ($h=0$)	-\$566.2	**
Medicaid PE: Severe Drought (lag 1)	-\$524.5	***
Medicare PE: Severe Drought (lag 1)	-\$144.3	***
Medicare PE: High CDD (lag 1)	+\$144.7	**
Medicare PE: High CDD (lag 2)	+\$137.2	***
Medicare PE: High HDD (lag 1)	-\$110.1	*
Medicare PE: High HDD (lag 2)	-\$192.7	***

Headline: Extreme Drought lag-2 and Cold Shock lag-1 raise medical debt & premiums; drought depresses Medicaid; heat and cold move Medicare in opposite directions.

County-Level Findings

Cold raises debt and premiums; heat depresses income;
hospital bad debt rises with cold and lagged drought

County-Level: Channels Mirror the State Findings



Cold channel:

- Raises debt prevalence & severity
- Raises premiums on impact
- **Raises** HH income (energy mix)

Heat channel:

- Raises debt median (lagged)
- **Reduces** HH income contemporaneously
- Raises premiums with 1–2 yr lag

Cross-level alignment:

- Cold → debt: both levels confirm
- Heat → income: county only
- Premium channels align in direction

County: Headline Coefficients and Hospital Channel

Two-way FE (fips + Year); state-clustered (RA-cluster for premiums); ** $p < 0.05$, *** $p < 0.01$

Premiums, debt, income (PW)

Coefficient	Estimate	
Benchmark Silver: High HDD ($h=0$)	+\$21.9	***
Benchmark Silver: High CDD (lag 2)	+\$35.3	***
Debt Median (PW): High HDD ($h=0$)	+\$31.2	**
Med HH Income: Z_Temp ($h=0$)	-\$88	***
Med HH Income: High HDD ($h=0$)	+\$173	***
PCPI: PDSI lag 2	-\$113	**
PCPI: RA-spec PDSI ($h=0$)	-\$206	***

Hospital bad debt per capita

Coefficient	Estimate	
High HDD ($h=0$)	+\$5.33	***
Z_Temp lag 2 (UW)	-\$2.41	***
Z_Temp lag 2 (PW)	-\$2.08	***
PDSI lag 2 (PW)	+\$0.50	*
Uninsured Rate (control)	+\$151	***

Cold raises hospital bad debt on impact; heat lag reduces it (composition / survivor effect).
Uninsured rate dominates the control block.

Weather-Swing (Delta) Analysis

Year-over-year climate volatility as a new source of variation;
new findings the level specs missed

Weather Swings: AQI Ratchet, Drought-PCPI, Temp-Debt

Δ FE primary spec; LP horizons $h = 0 \dots 3$; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Selected swing coefficients

Coefficient	Estimate	
Δ HDD $\rightarrow$ Silver Prem ($h=0$)	+0.033	***
HDD Onset $\rightarrow$ Silver Prem	+\$49.5	**
Δ Z_Temp $\rightarrow$ Debt Median ($h=3$)	+\$29.7	***
Δ Max_AQI $\rightarrow$ Hosp Bad Debt ($h=0$)	+0.005	**
Δ Max_AQI $\rightarrow$ Hosp Bad Debt ($h=3$)	+0.016	***
Δ Z_Precip $\rightarrow$ PCPI ($h=1$)	-\$274	***
Δ PDSI $\rightarrow$ PCPI ($h=1$)	-\$226	***

What the delta analysis adds

- **AQI ratchet** ($h=3$ effect $3 \times h=0$) — level specs were null; volatility identifies cost
- **Drought worsening** drives PCPI loss; recovery asymmetrically weaker
- **Temp swing** accumulates into medical debt over 3 years
- **HDD onset pricing**: first-year cold raises Silver Premium $\sim$ \$49

Key insight: weather *volatility* carries independent health-finance costs beyond level effects.

Cross-Level Synthesis

State, county, and weather-swing analyses tell a consistent story

Finding	State	County	Swing
Cold / HDD → medical debt ↑	✓	✓	✓
Cold / HDD → premiums ↑	✓	✓	✓
Cold / HDD → hospital bad debt ↑	—	✓	—
Drought → Medicaid ↓ / PCPI ↓	✓	✓	✓
Heat / CDD → HH income ↓ (lag)	—	✓	(asym.)
AQI → hospital bad debt ↑	(weak)	(weak)	✓ (new)
Heat vs. cold on <i>Medicare</i> (opposite signs)	✓	(partial)	—

Headline: cold shocks raise household financial burden; drought hits public payers and PCPI; weather *volatility* adds an independent channel (AQI ratchet; temp-debt accumulation).

Part 2 — April 2026 Committee Feedback Response

Random effects, post-exit dynamics, never-exposed DiD,
propagation evidence, humidity

What the Committee Asked

Five concerns across three domains, all addressed in this update except humidity

Econometric

- **Check random effects** as an alternative to FE
- **Lag / exit tests**: what happens when a county leaves a shock state?
- Find **natural experiments** for DiD — are there never-exposed counties?

General

- **Propagation pathways** must be backed by evidence (heat → delayed care; cold → utilization shifts)

Environmental

- Add **humidity** (PRISM tdmean) at state level

Response status

Concern	Status
Random effects	Done
Post-exit dynamics	Done
Never-exposed DiD	Done
Propagation pathways	Done
Humidity (PRISM)	Parked

Random Effects Is Rejected; Headlines Survive

Hausman tests on primary FE specs; matched plm two-way effects

Design

- `plm::plm(model="within"|"random",
effect="twoways"); phtest`
- 11 outcomes estimable (4 state, 7 county)
- 2 state outcomes failed RE fit on thin panels

Verdict

- **RE rejected for every estimable outcome at $p < 10^{-9}$**
- Headlines (Drought lag-2, Cold lag-1) have the **same sign** in RE

Selected Hausman results

Outcome	χ^2	p-value
<i>State</i>		
Medical Debt Share	233.6	4.5e-40
Total PC Health Exp	247.9	5.7e-43
Medicaid PE Health Exp	81.3	2.3e-10
Medicare PE Health Exp	834.4	2.1e-166
<i>County</i>		
Medical Debt Share	23,905	<1e-300
Benchmark Silver Real	23,545	<1e-300
PCPI Real	5,809	<1e-300
Civilian Employed	220.2	4.6e-41

Source: Analysis/random_effects_hausman.csv

Post-Exit Dynamics — What Happens When a County Leaves Shock State?

Local projections of *_Exit at $h = 0, 1, 2, 3$; answers the committee's exit-test question directly

Drought_Exit

- **PCPI partial scarring**: +\$1,044 p.c. at $h = 2$ ($p = 0.0002$)
- Recovery rebounds but does not fully restore pre-shock income
- Confirmed by Block B interaction (+\$964 at $h = 2$, $p = 0.010$)

HDD_Exit

- **Immediate hospital-cost relief**: -3.13 p.c. at $h = 0$ ($p = 0.014$)
- Sustained at $h = 2$: -2.88 ($p = 0.034$)
- Cold-shock cost is in the **exposure**, not lasting damage

CDD_Exit

- Persistent employment benefits after exit
- Likely part secular regional trend (caveated)
- Phase 2 LP rejects within-county heat scarring

Phase 0: Are There Never-Exposed Counties?

Inventory of zero-event counties over 2011–2023; feasibility memo selected primary cohorts

Never-exposed pool by shock

Shock	Total	Never-exposed	Share
Extreme Drought (PDSI ≤ -4)	3,225	2,534	78.6%
High HDD (top quintile)	3,225	2,303	71.4%
High CDD (top quintile)	3,225	2,121	65.8%
High AQI Max (>100)	3,225	100	3.1%

- Drought, HDD, CDD all have large never-exposed pools
- **AQI dropped from DiD**: 3.1% control pool too thin

Candidate sharp events

Event	Onsets	Onset rate
2012 Midwest Drought	139	4.4%
2022 Drought	110	3.5%
2013 HDD (cold)	407	13.0%
2018 CDD (heat)	250	8.0%
2023 AQI (wildfire smoke)	336	34.1%

Source:

Analysis/never_exposed_event_year.csv

2012 Midwest Drought: The Cleanest Natural Experiment

139 first-event counties (treatment) vs. 2,534 never-exposed (control), 2011–2023

Design

$$Y_{it} = \alpha_i + \gamma_t + \tau (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

- Treated: first-ever Extreme Drought event = 2012
- Control: never experienced Extreme Drought in panel window
- Cluster: state; SE: `feols(..., cluster="State")`

Why this is the cleanest piece in the thesis

- Within-county LP cannot rule out recurring-shock confounding
- 2x2 against never-exposed **closes that loophole**
- Drought 2012 is a documented historical event with well-defined geography

2x2 ATT results

Outcome	ATT	p
Medical Debt Share	−0.0062	0.49
Hosp Bad Debt PC	+6.26	0.29
PCPI Real	−\$1,311	0.027
Med HH Income Real	−\$992	0.24
Civilian Employed	−2,053	0.0001
Benchmark Silver Real	—	ACA collinearity

Source:

`Analysis/did/did_2x2_drought_2012.csv`

Callaway-Sant'Anna: Long-Run Cold-State Scarring

Manual CS estimator via `fixest` (R did package unavailable on local toolchain); never-treated controls

Drought (4 cohorts, 268 treated)

- PCPI at $e = 0$: **−\$1,050** ($p = 0.002$)
- Civilian Employed at $e = 0$: −142 ($p = 0.002$)
- Civilian Employed at $e = 11$: −3,915 ($p = 0.0001$)

HDD (2 cohorts, 230 treated)

- Civilian Employed at $e = 10$: **−4,982** ($p = 0.003$)
- Medical Debt Share at $e = 10$: **+4.9 pp** ($p = 0.0002$)
- Cumulative scarring through $e = 6$ –10

CDD: noisy, regional confounds

- Treatment concentrated in TX/GA/MS/FL
- Controls dominated by northern states
- Signs reverse across horizons — **de-emphasize**

Output

- Analysis/did/did_cs_att_gt.csv (427 ATT(g,t) cells)
- did_cs_event_time.csv (cohort-weighted profiles)

LP and DiD Are Complementary, Not Contradictory

The cold-scarring puzzle resolves once we see both designs measure different things

Local Projections / Exit-LP

- Estimand: within-county response when shock state changes year-over-year
- HDD_Exit $\rightarrow$ hospital-cost relief at $h = 0$
- **Read:** no within-county cold scarring after recovery

CS-DiD with never-exposed

- Estimand: persistent gap between ever-treated and never-treated populations
- HDD long-run: $-4,982$ jobs and $+4.9\text{pp}$ debt share at $e = 10$
- **Read:** treated counties end up worse off long-run

Reconciliation: cold-shock counties *recover* when shocks end (LP), but they *spend more years in shock state* than never-exposed counties do, so the **population gap accumulates** (CS-DiD).
Both are correct; they measure different things.

Propagation Pathways — Now Evidence-Backed

Each headline channel anchored to 2–4 peer-reviewed references with direction and time-scale

Pathway	Time-scale claim	Where identified	Strength
Heat → delayed care	Within-year deferral; lag-1/2 utilization	County Spec 2, CS-DiD CDD	Moderate
Cold → utilization shift	Contemporaneous + lag-1; long-run scarring at $e \geq 8$	State cold_shock_lag1, CS-DiD HDD	Strong
Drought → income → debt	Lag-1, lag-2 income; debt follows	State drought_lag2, 2012 DiD, Exit-LP recovery	Strongest
AQI → respiratory / cardiac	Within-year, episodic spikes	County continuous AQI, event-study High_AQI_Max, swing analysis	Limited

Selected references

- **Drought → income:** Hornbeck (2012, AER); Burke-Hsiang-Miguel (2015, Nature); Carleton et al. (2022, QJE)
- **Cold:** Deschênes-Moretti (2009, REStat); Gasparrini et al. (2015, Lancet); Andrews et al. (2017)

Humidity Integration — Status and Resumption Plan

Phase 4 deferred; the highest-priority residual gap from committee feedback

What we know

- Endpoint:
`services.nacse.org/prism/data/get/us/4km/needs/crra/mf`
- Returns zipped BIL raster (HTTP 200 confirmed)
- No state-aggregate API; must aggregate locally
- ~408 monthly rasters, ~1.5 GB on disk

Blocker

- Raster → state polygon aggregation
- None installed in local R 4.2.2; Windows install pulls heavy system deps (GDAL/GEOS/PROJ)

Resumption plan

- Build
`download_prism_humidity.R`,
`process_state_humidity.R`
- Integrate into state pipeline; re-run `run_analysis.R`
- Test whether **drought lag-2** and

The Analysis Now Has Two Pieces of Stronger Identification

And one genuine puzzle resolved by treating LP and DiD as complementary

New evidence

- **2012 Drought 2x2** is now the cleanest single identification piece (PCPI $-\$1,311$; Employed $-2,053$)
- **HDD long-run scarring** ($-4,982$ jobs at $e = 10$; $+4.9\text{pp}$ debt share at $e = 10$) was not visible in the within-county LP framework
- Drought_Exit shows **partial income scarring** — recovery rebounds but does not fully restore

Resolved puzzle

- Cold counties **recover on exit** (no within-county scarring)
- But the ever-treated population **ends up worse off** long-run (DiD)
- Both findings are correct; they measure different things

Methodological closure

- Random effects rejected; FE is the right spec
- AQI exclusion from DiD now defensible (3.1% never-exposed)

Open Caveats and Residual Risks

What the committee should know is still not addressed

Humidity confounding

- PRISM `tdmean` not yet a control
- Headline drought lag-2 and cold lag-1 effects **have not been stress-tested** against humidity
- Mitigation: Phase 4 resumption plan in place

CDD CS-DiD region confound

- Treatment concentrated in southern states; controls in north
- Even two-way FE leaves regional secular trends
- Reported as suggestive;

Multiple testing

- 427 ATT(*g,t*) cells $\times$ 7 outcomes $\times$ up to 11 event-times
- Headline cells survive Bonferroni within-shock
- Smaller-cohort cells do not — flagged in synthesis

County-master deduping

- Master is not strict one-row-per-(fips, year) due to rating-area joins
- Deduped in every committee-feedback script

Thank you.

Discussion and questions

Track artifacts at

`conductor/tracks/committee_feedback_april_2026/`